

Waffle Industries Forex Risk Management

Waffle Industries, a U.S. based dog toy manufacturer, has been experiencing some challenges lately due to increased competition in the industry and a downturn in the local economy. The firm's CEO and owner, Samantha Wolf, has been working hard to find new clients and boost profitability. Given the state of the industry in the U.S., she decided to explore opportunities overseas and develop an export market. After contacting many distributors and pet supply companies in Europe and Asia, she was finally able to land a deal with a large pet store chain based in the UK – PlanetPet!

PlanetPet expressed interest in Waffle's new and engaging dog toy line and wanted to place an order provided that costs were reasonable, and invoicing was done in British pounds. Samantha Wolf, eager to use idle capacity and pursue such a great opportunity, prepared and presented an offer right away. An agreement with PlanetPet was finalized last week and will be signed **today, April 20th**, with the following terms:

- Waffle Industries will supply 1 million units of their product to PlanetPet – 500k units to be shipped in a month and the rest to be shipped in 3 months.
- PlanetPet will make the following payments (in British pounds) to Waffle Industries:
 - £ 500,000 as cash down payment made **today**.
 - £ 700,000 to be paid in 1 month, at the time of the first shipment (**May 20th**)
 - £ 1,500,000 to be paid in 3 months, by which time the order would be complete (**July 20th**)

Ms. Wolf arrived at the total price of £2.7 million pounds by first pricing the order in dollars in March, when she was preparing the offer, and then converting that price to British pounds based on the spot exchange rate on that day – \$1.39/£. She estimated the costs on the assumption that export costs would run nearly the same as for existing domestic sales. This morning, she noted that the pound had somewhat weakened against the dollar - the exchange rate was at \$1.38 per pound. At that rate, the order of £2.7 million pounds was worth \$3,726,000 or \$27,000 less than when the bid was made. To secure the deal with PlanetPet, Ms. Wolf had offered a very competitive price and had not built in a very large profit margin.

Concerned about the pound depreciating even further over the next three months, and the dollar-value of the two expected payments declining significantly, Ms. Wolf decided to consult her bank and ask for advice on managing foreign exchange risk. Your team was assigned the task of helping Ms. Wolf manage her forex exposure and ensure the continued success of Waffle Industries.

Your task is to prepare a consulting report for Ms. Wolf, outlining and analyzing in detail the four strategies that are available to her (listed below). Compare and contrast the different choices, pointing out advantages and disadvantages of each, and make a recommendation for the best course of action.

Ms. Wolf is interested in the effective price (in \$) that she is expected to receive for the order with each strategy. While the precise outcome cannot be determined for all four strategies, your team should use the data provided below to forecast future exchange rates and estimate what the dollar-denominated cash inflows would be under different scenarios (e.g. low, high, most likely estimate).¹ In your analysis, make any reasonable assumptions that you need.

Evaluate and compare all strategies listed below:

1. Do nothing (remain unhedged)
2. Hedge the exposure in the forward exchange market
3. Hedge the exposure in the futures market
4. Hedge the exposure in the options market

¹ There are several ways to forecast the expected spot rate for May 20 and July 20. You could run a regression with time trend and forecast it. Or you could use a moving average approach. Or you could use a probability distribution approach to forecast the future spot rates on those days.

Data for the analysis:

- Exchange rates over the last three months (in \$ per pound):

Date	Spot Exchange rate (in \$/£)
January 20	1.365122
January 21	1.371789
January 22	1.368167
January 23	1.36816
January 24	1.36816
January 25	1.366316
January 26	1.372636
January 27	1.369854
January 28	1.371808
January 29	1.371836
January 30	1.37125
January 31	1.3705
February 01	1.367998
February 02	1.364359
February 03	1.365495
February 04	1.366391
February 05	1.372448
February 06	1.37365
February 07	1.37365
February 08	1.373825
February 09	1.379453
February 10	1.384783
February 11	1.381693
February 12	1.385397
February 13	1.3842
February 14	1.3842
February 15	1.391798
February 16	1.390695
February 17	1.38438
February 18	1.395345
February 19	1.401768
February 20	1.4015
February 21	1.40165
February 22	1.406955
February 23	1.409652
February 24	1.409652
February 25	1.410686
February 26	1.395469
February 27	1.3925
February 28	1.3925

March 01	1.394224
March 02	1.395255
March 03	1.398308
March 04	1.400864
March 05	1.379777
March 08	1.380462
March 09	1.388667
March 10	1.388985
March 11	1.397868
March 12	1.39205
March 14	1.39245
March 15	1.388648
March 16	1.389265
March 17	1.388021
March 18	1.393544
March 19	1.387575
March 20	1.387
March 21	1.387
March 22	1.386203
March 23	1.377591
March 24	1.37207
March 25	1.373315
March 26	1.379175
March 27	1.3791
March 28	1.3791
March 29	1.377818
March 30	1.372034
March 31	1.380167
April 01	1.382103
April 02	1.382074
April 05	1.390337
April 06	1.382074
April 07	1.374089
April 08	1.3743
April 09	1.37258
April 10	1.37085
April 11	1.37085
April 12	1.374448
April 13	1.373806
April 14	1.379977
April 15	1.378578
April 16	1.382294
April 17	1.38335
April 18	1.38335

- Forward quotes (from the derivative trading desk of your bank)
The trading desk of your bank just quoted the following forward exchange rates (in \$/£)

Expiration	Forward Price
May 20 th	1.385
July 20 th	1.395

- Futures quotes (from CME) – Each contract is on 62,500 British pounds, in \$/£

Expiration	Futures Price
May	1.384
June	1.387
July	1.393
August	1.399

- Option quotes (pick only one strike price for your analysis)
Your team contacted your bank's derivative trading desk and would offer an option contract specific to Ms. Wolf's needs. The quotes are the following:

Amount £700,000			Amount £1,500,000	
Expiration date: May 20			Expiration date: July 20	
Strike price	Call	Put	Call	Put
\$1.380	\$0.018/£	\$0.014/£	\$0.035/£	\$0.021/£
\$1.385	\$0.016/£	\$0.016/£	\$0.032/£	\$0.023/£
\$1.390	\$0.013/£	\$0.019/£	\$0.029/£	\$0.026/£

Other Considerations: Ms. Wolf converted the down payment of £ 500,000 at the spot rate of \$1.38/£ and now has \$690,000. These funds will be used to purchase inputs for the manufacturing process and to finance working capital needs. The issue of working capital is important but not the main focus of this case. Discuss it if it helps your analysis.